

Thought Experiments through Chapter 5 of *Foundations of Quantum Theory*

—From the Wall of Identification in Measurement to the Rectangular Phase-Energy Window

Author: Noriaki Kihara, WF System Co., Ltd., ORCID: 0009-0004-6753-4020 **Version:** v1.0 (incorporating multiple rounds of AI peer review: Gemini $\times$ 1, Grok $\times$ 1, ChatGPT $\times$ 2) **Date:** May 2026 **Concept DOI:** 10.5281/zenodo.20398526 **v1 DOI:** 10.5281/zenodo.20398527 **Previous paper:** *Thought Experiments through Chapter 3 of Foundations of Quantum Theory* v3.0.1 (Concept DOI: 10.5281/zenodo.20391522; v3 DOI: 10.5281/zenodo.20393018) **License:** CC BY 4.0

Abstract

Against the background of complex Hilbert space, observables, measurement, uncertainty relations, the one-dimensional particle, the box potential, and the tunneling effect developed in Chapters 1–5 of Shimizu Akira’s *Foundations of Quantum Theory, New Edition* [1], we organize, through seven staged thought experiments, measurement precision, uncertainty relations, wave packets, quantum correlations, the algebra of observables, the ontology of physical quantities, and the finite-width structure of particles into a unified picture. Carrying over the five thought experiments of the previous paper (*Thought Experiments through Chapter 3* v3.0.1), this paper adds, as the region corresponding to Chapters 4–5, the representation of physical quantities over a complex phase space (Thought Experiment VI) and the rereading of particles as rectangular phase-energy windows with a central phase and a finite width (Thought Experiment VII). The starting point is the “wall of identification” in classical metrology; the terminal point is the ontological rereading “a particle is a rectangular phase-energy window with a central position phase and a finite width, and the observed wave form appears as the Fourier low-order partial sum of that window (the low-passed image in the observation bandwidth).” The core reading of this paper is the semiclassical/geometric interpretation that positions phase-space invariants as symplectic capacities, represented by de Gosson’s quantum blobs [9]. This paper does not modify the mathematical predictions of standard quantum theory; within the scope of one-dimensional quantum mechanics, measurement, uncertainty, and the box potential treated in Chapters 1–5, it offers a conceptual rereading consistent with the existing calculational formalism. For regions beyond Chapters 1–5 — Lorentz covariance, locality of fields, positivity, unitarity, scattering amplitudes, spin statistics, gauge symmetry, particle creation and annihilation — whether the present rereading extends consistently is reserved as an independent task. The paper does not deny the Born rule. Rather, it treats the position-phase window $P_x(\theta)$ as a shape function, not a probability density. When the observation system reads this shape function as the wave function corresponding to a detection basis, the standard inner-product rule $p(a) = |\langle \varphi_a | \psi \rangle|^2$ then yields the probability density.

Thought Experiment I: The Wall of Identification in Metrology

Setup

Let a physical quantity A take real values. Using a distance-concept-based ruler (returning only discrete values with minimum spacing Δ), we measure the distance L between two points A_1, A_2 .

In each measurement, one of the two adjacent grid points $\{k\Delta, (k+1)\Delta\}$ enclosing the true value is returned, with probabilities determined by the in-cell position r of the true value:

$$P(\text{upper}) = r/\Delta, \quad P(\text{lower}) = 1 - r/\Delta$$

(position-dependent dithering).

Behavior of the N-fold average

When $A_1 = A_2$, the single-measurement distance $D = \varepsilon_2 - \varepsilon_1$ takes values $\{-\Delta, 0, +\Delta\}$. Writing the in-cell position of the true value as $q = r/\Delta \in [0, 1]$, in general:

- $P(D = +\Delta) = q(1 - q)$
- $P(D = -\Delta) = q(1 - q)$
- $P(D = 0) = q^2 + (1 - q)^2$

Hence $E[D] = 0$, $\text{Var}[D] = 2q(1 - q)\Delta^2$, $\sigma_D = \sqrt{2q(1 - q)} \cdot \Delta$.

In particular, at the symmetric position $q = 1/2$ the probability distribution becomes $\{1/4, 1/2, 1/4\}$, with $\text{Var}[D] = \Delta^2/2$ and $\sigma_N = \Delta/\sqrt{2N}$.

As $N \rightarrow \infty$, $\sigma_N \rightarrow 0$ and $\hat{L}_N \rightarrow L$ with probability 1.

Condition for distinguishability

To recognize A_1 and A_2 as two distinct objects, both must belong to different Δ -cells. If they fall within the same cell, the ruler cannot tell “one point or two,” and the quantity “distance between two points” does not arise.

The condition for the distance-measurement problem to be well-posed is therefore

$$L \geq \Delta.$$

The region $L < \Delta$ is not “buried in measurement error” but a region in which the very concept of “two points” is undefined.

Conclusion

When measuring distance with a distance-concept-based ruler, any metric below the ruler spacing Δ is **measurement-undefined** even before invoking the uncertainty relation. Δ acts not as a resolution but as the threshold of individuation.

Within the range $L \geq \Delta$, infinitely many measurements can attain arbitrary precision regardless of the absolute value of Δ .

This argument holds for any distance-concept-based ruler and is structurally analogous to, but independent of, the minimum-length discussion in quantum gravity at the Planck scale [10].

The relation between epistemic and ontological limits: The “undefined” above is not the ontological claim that distances below Δ **do not exist** (that space is discretized). It is the claim that, viewed from inside an observation system built on a ruler of spacing Δ , two points are **epistemically indistinguishable, and within that observation system there is no operational meaning to treating them as anything other than ontologically identical** — that is, an epistemic limit manifests as a geometric one. Thus our position is not to place epistemology and ontology side by side as independent, but to take the limit of distinctions constructible inside the observation system to be the geometric limit of individuation: an epistemology-led geometry. From this viewpoint, the external question “is the real space discrete or continuous?” is not entered.

More precisely: Rather than the ontological phrasing “two points do not exist below Δ ,” the formulation “no projection operator distinguishes two points in the observational algebra generated by a Δ -spaced ruler” is more appropriate as a connection to quantum measurement theory. This discussion can be made consistent with the framework of generalized measurements via POVMs / Kraus operators, and positioned as a special case of coarse-grained measurement.

Note that when A_1, A_2 are externally pre-labeled as distinct objects, sub-cell precision can be estimated from the mean difference via repeated measurement ($N \rightarrow$ convergence by position-dependent dithering).

“Undefined” here refers to the case where no operation exists, inside the observation system, to individuate the two objects.

Thought Experiment II: The Locus of Fluctuation

Setup

Consider the reverse situation. The ruler has infinite precision and returns the input value as-is. On the other hand, the physical quantity A itself fluctuates around its true value A' as

$$A_i = A' + \xi_i, \xi_i \sim \text{Uniform}[-\Delta/2, +\Delta/2].$$

Assume $L > \Delta$.

N-fold average

- $E[\hat{L}_N] = A'$
- $\text{Var}[L_i] = \Delta^2/12$
- $\sigma_N = \Delta/\sqrt{12N} \rightarrow 0$

Comparison with Thought Experiment I

Concretizing both cases with true value 7.7 and spacing $\Delta = 1$, both yield the same empirical distribution: “in 100 measurements, 7 appears about 30 times, 8 about 70 times, and the mean converges to 7.7.”

	Fluctuation on the ruler side	Fluctuation on the physical-quantity side
Single-output space	Discrete $\{k\Delta, (k+1)\Delta\}$	Continuous
Probability distribution	Position-dependent Bernoulli	Sum of uniforms
$N \rightarrow \infty$ limit	True value	True value
Distinguishable from data	Impossible	

Two-sided fluctuation

In the general case where physical-side fluctuation ξ (width δ) and ruler-side fluctuation η (width σ) coexist independently:

- Observed value $L_i = A' + \xi_i + \eta_i$
- $\sigma_N = \sqrt{\text{Var}[\xi] + \text{Var}[\eta]}/\sqrt{N} \rightarrow 0$

The $1/\sqrt{N}$ decay is preserved, but separating δ and σ from a single observed series requires additional assumptions outside the observation (independent calibration of the ruler, time-series decomposition, prior knowledge of distributional form, etc.).

Conclusion

Define the observed magnitude of fluctuation as Δ_{obs} . Whether it comes from the quantization width of the ruler, from the fluctuation of the physical quantity itself, or from their combination, cannot be decided from the observed data alone.

What can be stated rigorously is not “convergence to an objective true value” but “convergence to an effective true value defined by the observer,” together with the $1/\sqrt{N}$ decay.

More precisely: “Cannot be decided from observed data alone” means “under the conditions of a single observation series, a single measurement setting, and no prior calibration.” When the apparatus Hamiltonian is known, it is theoretically possible to distinguish system-side fluctuation from environment- (apparatus-)

side fluctuation via decoherence theory for open quantum systems. The claim of this paper is restricted to non-identifiability from observation data alone, and connects to epistemic aspects (QBist interpretation) that depend on the observer's knowledge state.

Thought Experiment III: Wavelength Representation of Momentum and the Uncertainty Relation

Setup

Extend the object of Thought Experiment II to a quantum wave packet. For position x , consider the center of the wave packet (expectation value $\langle x \rangle$) x_1 and the spread δx . Here the “center” is a label of the wave packet's distribution; we do not take the stance that there is an objective true value outside the observation.

Similarly, for momentum p , consider the center (expectation value $\langle p \rangle$) p_1 and the spread δp .

Rewriting via wavelength

The de Broglie relation [2]:

$$p = \frac{h}{\lambda} = \hbar k, \quad k = \frac{2\pi}{\lambda}$$

gives the fluctuation of momentum as

$$|\delta p| = \frac{h}{\lambda^2} |\delta \lambda| = \hbar |\delta k|$$

(the differential relation $dp/d\lambda = -h/\lambda^2$ has its sign absorbed in taking absolute values for the fluctuation width).

Rewriting the uncertainty relation $\Delta x \Delta p \geq \hbar/2$ [7] in wave-number form:

$$\Delta x \cdot \Delta k \geq \frac{1}{2}.$$

Observation

In this form, Planck's constant $\hbar$ disappears from both sides of the inequality. What remains is the mathematical Fourier inequality on the product of position and wave-number.

Planck's constant $\hbar$ functions as a dimensional conversion factor connecting units of position (m) and momentum (kg · m/s). A similar view, treating Planck's constant not as an observable but as a unit-choice convention, has been presented by Ralston [8].

Qualitatively:

- Determining position precisely narrowing the position wave packet broadening the wavelength distribution momentum becomes vague
- Determining momentum precisely narrowing to a single wavelength spreading throughout space position becomes vague

Form of the wave packet	Spatial extent	Wavelength distribution
Delta function	$\delta x = 0$	All wavelengths
Rectangular pulse	Finite	Sinc function
Pure sine wave	Whole space	Single wavelength

Conclusion

In the wave-number representation, the position-momentum uncertainty relation becomes $\Delta x \Delta k \geq 1/2$, the lower bound on the product of conjugate quantities in the Fourier transform. Planck's constant appears as a dimensional conversion factor.

More precisely: $\hbar$ is the action quantum appearing in the canonical commutation relation $[x, p] = i\hbar$; this physical content is independent of the fact that it can be set to $\hbar = 1$ in natural units. The claim here is restricted to the rewriting: in the wave-number representation the canonical commutation relation becomes $[x, k] = i$, and the uncertainty relation can be reread as the geometric inequality of wave-number conjugacy. “ $\hbar$ disappears” means it disappears from the surface of the description, not that the physical role of $\hbar$ as a quantum of action is denied.

Thought Experiment IV: Quantum Correlation as a Composite Wave Packet

Setup

Consider a state localized in two spatial regions A, B and correlated via a common conserved quantity (momentum, energy, etc.). This is the system standardly described as a “two-particle entangled state” [3, 4, 5].

We attempt to describe it as “a single composite wave packet localized in two regions in space.”

The string metaphor

A long, tensioned string vibrating on a baseline. If one end is suddenly fixed, a soliton-like deformation appears at the opposite end. This is the consequence of the whole string obeying conservation laws as one system; no information has propagated from one end to the other.

Geometry of the composite wave packet

- A composite wave packet occupies a single area element in phase space.
- In space, it has two localized peaks in regions A, B .
- A local interaction on side A shrinks Δx_A .
- By the Robertson inequality [6] and Robertson–Schrödinger-type covariance constraints, when the spread along the projection direction on side A shrinks, a compensatory change appears in the conjugate direction or in covariance components, broadening Δp_A .
- To satisfy the geometric constraint on the whole composite wave packet — the symplectic capacity — the spread of side B 's wave packet also deforms geometrically.

However, the deformation on side B is a change in the description of the post-state conditioned on side- A 's measurement outcome; the unconditioned reduced density matrix of side B alone does not change under local operations on side A . This distinction is consistent with the no-signalling theorem of standard quantum theory.

In this description, no information transmission from A to B is introduced. The observed correlations are a geometric consequence reflecting the state of the whole system together with conservation laws, symmetries, and symplectic capacity constraints.

The geometric invariant

What constrains the geometry of the composite system is not individual wavelengths or positions but the **symplectic capacity** formed by the product of the spreads of conjugate observables (the area element in phase space):

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}, \quad \Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$$

The left-hand side has the dimension of area, and the inequality denotes the lower bound on area. This is not a simple conservation law for $\Delta x \Delta p$ but a geometric constraint by the symplectic capacity invariant under canonical transformations.

Strictly speaking, in multi-degree-of-freedom systems the symplectic capacity is not defined as the total phase-space volume or as the product of individual uncertainty products, but as a symplectic invariant for projections onto canonical 2-planes (Gromov’s non-squeezing theorem [15]). We use the $\Delta x \Delta p$ -type area element as an intuitive shadow conveying the essence.

Conclusion

Quantum correlations can be understood through the geometric picture “reading a single composite state as projections onto multiple regions in space.” The observed correlations are a geometric consequence reflecting the state of the whole system, conservation laws and symmetries, and symplectic capacity constraints; this description requires no information transmission between spatially separated systems.

More precisely: The composite-wave-packet picture in this paper is an attempt to read a single state vector on a tensor-product Hilbert space (in standard quantum theory) through the geometric intuition of phase space. The soliton-string analogy is an intuitive presentation of the geometric structure of conservation laws and is not a complete re-derivation of quantitative predictions such as the maximum Bell-inequality violation $2\sqrt{2}$ (the Tsirelson bound) of the CHSH inequality. Those quantitative predictions are derived from the algebraic structure of quantum theory independently of the classical wave analogy. The “no information transmission” point is consistent with the no-signalling theorem of standard quantum theory (partial trace of density operators, invariance under local operations).

Thought Experiment V: The Algebra of Observables

Preface on the scope of the metaphor: This section uses intuitive terms such as “wave packet,” “area element,” and “projection.” These serve as valid metaphors for the semiclassical/geometric interpretation of pure states in the position–momentum system (infinite-dimensional, continuous spectrum). In spin systems or finite-dimensional systems, the wave packet must be reread as a “point on projective Hilbert space, a state direction,” and the area element as the “geometry of the non-commutative algebra generated by projection operators.” Each thought experiment’s closing “More precisely” section supplies the rigorous mathematical correspondence.

Setup

Enumerate several pairs of physical quantities obeying the uncertainty relation $\Delta A \Delta B \geq | \langle [A, B] \rangle | / 2$:

- Position x momentum p
- Spin S_x S_y S_z
- Polarization (H/V D/A R/L)
- Angular momentum L_x L_y L_z
- Energy time (note: the energy-time uncertainty has a different mathematical status from Robertson-type uncertainties between position-momentum or spin components, since time in ordinary quantum mechanics is not defined as a self-adjoint operator)

All of these share a common structure.

Common structure

Pair of observables	Shared structure	Role of distinct operators
$x \ p$	Area element in phase space	Direct coordinate Fourier coordinate
$S_x \ S_y \ S_z$	Area element in spin phase space	Projections onto different axes
Polarization bases	Area element on the polarization plane	Projections onto different bases
$L_x \ L_y \ L_z$	Angular-momentum phase space	Projections onto different axes
$E \ t$	Energy-time phase space	Different representations

Unified picture

In complex Hilbert space:

Formal object	Geometric interpretation
Eigenvector $ \psi\rangle$	Wave packet (a single area element in phase space)
Operator $\hat{A}$	Projection extracting a real value from the area element
Eigenvalue a	The result of the projection
Non-commutativity $[\hat{A}, \hat{B}] \neq 0$	Two projections viewing the same area element from different directions

Conclusion

Observables obeying an uncertainty relation are described as different projections of the same wave packet (eigenvector). The product of non-commuting projections has the area element of the wave packet as a lower bound (Robertson inequality).

In this paper’s geometric reading, observables are understood as quantities that extract real-valued distributions from states, with values obtained through measurement bases or spectral projections. The wave packet, in the semiclassical phase-space representation, is interpreted as an area element, whose spread is restricted by the Robertson–Schrödinger-type covariance constraint and by the symplectic capacity. The projections of observables do not take arbitrary independent values: they are correlated under the geometric invariant of the same state.

The area-element picture of this paper is consistent with de Gosson’s [9] quantum blob (the minimum unit of phase space corresponding to the Robertson–Schrödinger minimum uncertainty, measured as a symplectic capacity — an area-type invariant on canonical 2-planes). This paper is positioned as a reconstruction of de Gosson’s mathematically heavy symplectic-geometric framework in the intuitive form of a thought experiment as a reading of Chapter 1 of Shimizu’s textbook. Related geometric formulations include the Ashtekar–Schilling formulation based on the Kähler structure of projective Hilbert space [11] and the algebraic approach treating the algebra of observables as primary [12].

More precisely: “Operator = projection” can be confused with the specific notion of a projection operator. It is more appropriate to say that observables are self-adjoint operators, and in projective measurements values are extracted via spectral projections (in generalized measurements, POVMs). “Eigenvector = wave packet” is a useful metaphor for position-momentum systems; in spin systems or finite-dimensional systems, it should be read as a “state direction, a point on projective Hilbert space.” The phase-space area-element picture in this paper is a semiclassical/geometric interpretation; rigorous purely-Hilbert-space formulations are provided by Weyl quantization, the Wigner function [13], the Moyal bracket [14], and geometric quantization.

Note also that the Robertson inequality itself is a lower-bound condition; there is no general absolute upper bound on $(\Delta x)(\Delta p)$ for position-momentum in infinite-dimensional Hilbert space (for a free particle’s Gaussian wave packet, Δx can increase with time). However, in finite-dimensional systems, with bounded observables, or in state-dependent contexts, upper-bound-type uncertainty relations — “reverse uncertainty relations” —

have been studied in recent years. The claim of this paper about geometric invariants is most accurately read not as a simple conservation law for $\Delta x \Delta p$ but as the invariance of the symplectic capacity under canonical transformations (de Gosson’s quantum blob [9]).

Thought Experiment VI: Are Physical Quantities Real Numbers? — Physical Quantities as Quantities over a Complex Phase Space

Terminology note: on “complex phase space”

By “complex phase space” we mean not a single standardized term but a **working umbrella term for complex-phase structures** appearing commonly across complex Hilbert space $\mathcal{H}$, the Kähler structure of projective Hilbert space $\mathbb{P}(\mathcal{H})$, the Wigner–Moyal phase space, and the phase factor $e^{i\theta}$. The strict mathematical object referred to differs by context, but the common feature — “real projections back a complex-phase structure” — is captured by this convenient name.

Setup

Through Thought Experiments I–V, we reached an algebraic picture in which observables are described as different projections of the same wave packet. Here we invert the question: what is the substance of a physical quantity *before* it is projected? Does the fact that observed values are real numbers imply that the underlying structure of physical quantities is closed over the real-number field?

Three assumptions

We assume the following.

Assumption 1: Observed physical quantities are real-valued. However, this is a constraint on the output type of the observation operation and does not imply that the underlying structure generating the quantity is closed over the reals. Just as wave functions are complex while observed probabilities are real, even if what is observed is real, we allow the underlying generating structure to be complex.

Assumption 2: Fundamental physical quantities are defined as quantities over a complex-valued phase space. Real-valued observed values appear as real projections of that complex-phase structure (real part, absolute value, functional value, etc.):

$$\mathcal{L} = \mathcal{L}(z, \bar{z}), \quad L_{\text{obs}} = F(\mathcal{L}, \bar{\mathcal{L}}) \in \mathbb{R}$$

Assumption 3: Physically stable values are selected not as continuous arbitrary values but as discrete values satisfying a phase-closure condition (the phase closes consistently across the system). One-dimensionally this is the standing-wave condition $L = n\lambda/2$; in phase space, it corresponds to the Bohr–Sommerfeld / EBK quantization condition $\oint p dq = 2\pi\hbar(n + \mu/4)$ for the action integral [16]; more generally, it corresponds to the integrality condition $[\omega]/(2\pi\hbar) \in H^2(M, \mathbb{Z})$ of geometric quantization. At boundaries and turning points, Maslov phase corrections of $\pi/2$ units appear.

Complex representation of the position phase

Representing position x as a phase $\theta_x = kx$, the central position x_0 corresponds to the central phase $\theta_x = kx_0$, and the spatial spread Δx corresponds to the phase width $\Delta\theta_x = k\Delta x$.

Figure VI-1: Complex representation of the position phase and phase-space image. The left panel shows, on the complex plane, the central phase $\theta_x = 180^\circ$ and the surrounding angular width $\pm\Delta\theta_x = 60^\circ$. $\Delta\theta_x$ is an angular width, not a radial spread. The right panel shows the same structure as a wave packet on the phase axis θ , with the real part of $\text{Re } \psi(\theta) = A(\theta) \cos(k(\theta - \theta_x) + \theta_x)$ as a thick line and the envelopes $\pm A(\theta)$ as dashed lines. The two panels are connected by the correspondence $\theta_x = kx_0$, $\Delta\theta_x = k\Delta x$.

Position as a complex phase with uncertainty

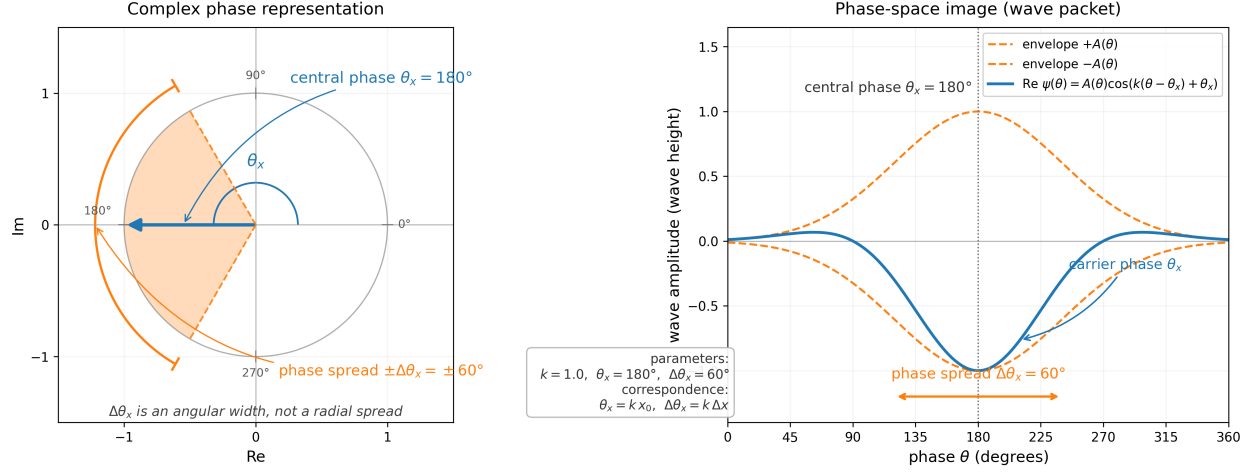


Figure 1: Complex representation of the position phase and phase-space image

Connection to existing theories

This model connects to existing quantum geometry:

- **Quantum geometric tensor** [18]: $Q_{ij} = g_{ij} + (i/2)\Omega_{ij}$, where the real part is a distance (Provost–Vallée / Fubini–Study metric) and the imaginary part is the Berry curvature (symplectic structure, non-commutativity).
- **Complex Hilbert space and projective space** [11]: physical states are treated as rays, and projective space $\mathbb{P}(\mathcal{H})$ naturally becomes a Kähler manifold.
- **Wigner–Moyal formalism** [13, 14]: quantum theory described as a pseudo-probability distribution on phase space.
- **Non-Hermitian / PT-symmetric quantum theory** [19]: self-adjointness of observables is not adopted as a foundational axiom; real observed values are recovered through an appropriate inner product and metric operator.
- **Ralston (2020)** [8]: Planck’s constant is treated not as an observable but as a unit-choice convention.

Conclusion

The substance of a physical quantity can be read not as a single value on the real line but as a phase-closure structure on a complex phase space. Under this assumption, uncertainty, interference, entanglement, and quantization are not separate phenomena but the same structure seen through different observation projections.

More precisely: This hypothesis is not a new predictive theory but an ontological/conceptual reinterpretation of existing quantum theory. The mathematical predictions of standard quantum theory are not changed. What breaks down when complex structure is placed on the side of substance is, e.g., the standard measurement theory based on self-adjointness of observables as a foundational axiom, complex-time QFT without reflection positivity, or a complexified standard model that fails to recover a unitary S-matrix [20]. These break-downs do not mean “complex structure itself is bad” but indicate that projection rules onto real observables, positivity, and unitarity must be recovered separately. The study by Renou et al. (2021) [21] on experimental distinction between real-number and complex-number versions of quantum theory points toward the direction of experimentally testing the physical necessity of complex structure.

Thought Experiment VII: Are Particles Rectangular Phase-Energy Windows? — A Rereading of the One-Dimensional Particle and the Box Potential

Setup

We perform the thought experiment of rereading the one-dimensional particle, the well potential, the box potential, and the tunneling effect treated in Chapter 5 of Shimizu’s *Foundations of Quantum Theory* (and similar one-dimensional quantum-mechanics textbooks) — not as the behavior of a particle under external conditions but as the foundational image of the particle itself.

Position phase as a rectangular phase window

In Thought Experiment VI the position phase was introduced as a quantity over a complex phase space. To give a more concrete form, we place the position phase as a rectangular phase window:

$$R_x(\theta) = \begin{cases} 1, & |\theta - \theta_x| \leq \frac{\Delta\theta_x}{2}, \\ 0, & |\theta - \theta_x| > \frac{\Delta\theta_x}{2}. \end{cases}$$

Here θ_x is the central phase and $\Delta\theta_x$ is the full width of the window. The value is 1 in the region $|\theta - \theta_x| \leq \Delta\theta_x/2$ and 0 elsewhere. This is the **body** of the particle / position phase.

The observation image as a Fourier partial sum

Expand the rectangular phase window $R_x(\theta)$ in a Fourier series of period 2π , and define the observation-bandwidth low-pass operator L_Λ as “the operation truncating at the N -th harmonic.” The observation image is

$$P_x^{\text{obs}}(\theta) = L_\Lambda[R_x(\theta)] = S_N(\theta) = \frac{a_0}{2} + \sum_{n=1}^N a_n \cos(n(\theta - \theta_x))$$

where the Fourier coefficients are (even expansion about the center θ_x):

$$a_0 = \frac{\Delta\theta_x}{\pi}, \quad a_n = \frac{2}{n\pi} \sin\left(\frac{n\Delta\theta_x}{2}\right) \quad (n \geq 1).$$

Figure VII-1: The body of the rectangular phase window $R_x(\theta)$ (left) and its observation image as Fourier partial sums $S_N(\theta)$ (right). The left panel shows the underlying form, the rectangular phase window $R_x(\theta)$ (center $\theta_x = 180^\circ$, full width $\Delta\theta_x = 120^\circ$). The right panel shows the partial sums $S_N(\theta)$ truncating the Fourier series at the N -th harmonic, for $N = 1, 3, 7$. For $\Delta\theta_x = 120^\circ = 2\pi/3$: $a_0/2 = 1/3$ (duty cycle), $a_1 \approx +0.55$, $a_2 \approx +0.28$, $a_3 = 0$ (zero of $\sin(n\pi/3)$), $a_4 \approx -0.14$, $a_5 \approx -0.11$, $a_7 \approx +0.08$. The low-pass operator L_Λ corresponds to truncating the Fourier series at the N -th harmonic; as N grows the partial sums converge to the rectangle (with Gibbs-type ringing at discontinuities). The lowest order S_1 consists of the DC term $a_0/2 = 1/3$ and the first cosine harmonic $a_1 \cos(\theta - \theta_x)$.

Strictly speaking, when treating an observation image of a rectangular window with bandwidth limitation, not only Fourier partial sums but also the prolate spheroidal wave functions (PSWF) of Slepian–Pollak [31] and Landau–Pollak [32] — which optimize the simultaneous concentration of time width and bandwidth — appear naturally. Moreover, by Hardy’s theorem [33], a function ψ and its Fourier transform $\hat{\psi}$ cannot both decay rapidly at once. Therefore the rectangular body, in a strict sense, can only be seen at observation as a band-limited effective waveform. The Fourier partial sum $S_N(\theta)$ of this paper is positioned as an intuitive, elementary approximation to these rigorous frameworks.

Rectangular phase-window body and its Fourier partial sums

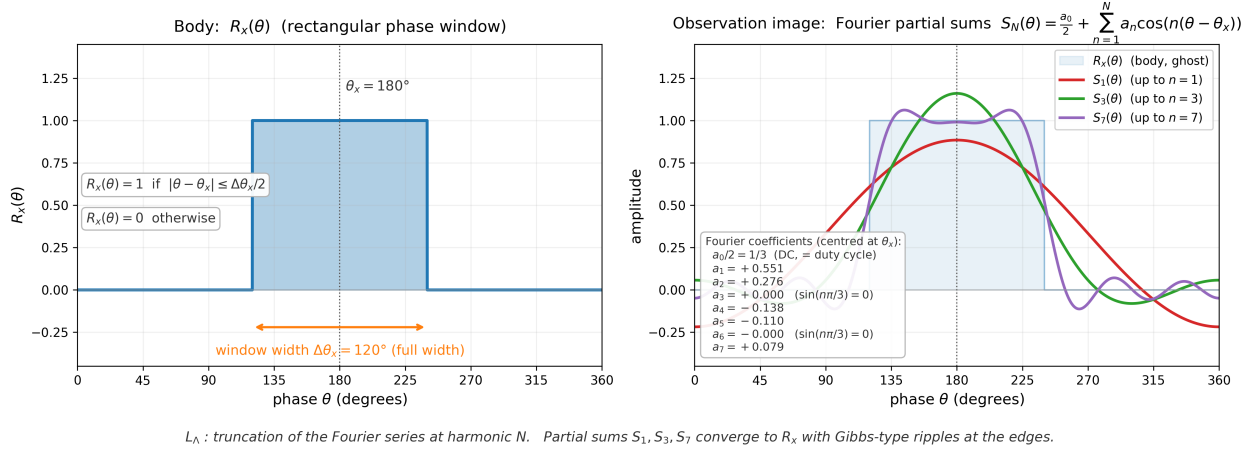


Figure 2: Body of the rectangular phase window and its Fourier partial sums

Definition of the particle (definitional hypothesis)

We redefine the particle as follows:

Particle = rectangular phase-energy window with central phase θ_x and finite width $\Delta\theta_x$

This is at present a **definitional hypothesis**. Why $R_x(\theta)$ is stable, by what dynamics the width $\Delta\theta_x$ is determined, why an energy density E_0 rides on it, and how observed scattering and binding energies match experimental values — none of this is derived within the scope of this paper. These are the questions answered, from specific Lagrangians, by the Skyrme [23], MIT bag [24], Friedberg–Lee–Sirlin [25], and Q-ball [26] lineages. The rectangular window of this paper is merely an alternative starting point for the same questions.

If we put an energy density on the rectangular window:

$$E_x(\theta) = E_0 R_x(\theta) = \begin{cases} E_0, & |\theta - \theta_x| \leq \frac{\Delta\theta_x}{2}, \\ 0, & |\theta - \theta_x| > \frac{\Delta\theta_x}{2}. \end{cases}$$

Geometric interpretation of the energy density E_0 : In standard quantum theory, energy is tied to the Hamiltonian and introduced as the generator of time evolution (phase rotation). The static spatial energy density $E_x(\theta)$ of this paper is consistently read not as Hamiltonian-energy itself but as an action-integral density over the phase window: the total action over the rectangular window is $S = \oint E_x(\theta) d\theta = E_0 \Delta\theta_x$, providing a phase-closure condition consistent with the Bohr–Sommerfeld / EBK quantization condition $\oint p dq = 2\pi\hbar(n + \mu/4)$ [16][17] introduced in Thought Experiment VI. That is, E_0 is not energy as the generator of time evolution but the expression of the action density that closes the rectangular window phase-coherently as a single quantum state. The consistency with Schrödinger-equation time evolution is beyond the scope of this paper and reserved as a follow-up task (the commutativity of the bandwidth-limitation operator giving $S_N(\theta)$ with the time-evolution operator).

This rereading gives:

Standard view	This paper's rereading
The particle is a point object	The particle is a rectangular phase-energy structure with finite width
The wave function $\psi(x)$ represents the particle's state	The observed wave form is the Fourier partial sum of the rectangular window under observation-bandwidth limitation
$ \psi(x) ^2$ is the probability density	The position-phase width itself carries the observation width. $P_x(\theta)$ is a shape function, not a probability density
Uncertainty arises from non-commutativity / measurement variance	The uncertainty $\Delta\theta_x$ is intrinsic to the definition of the particle
The integral of the probability density is 1	The normalization of the shape function is $P_x(\theta_x) = 1, P_x(\theta_x \pm \Delta\theta_x/2) = 0$

Ontological inversion

The central claim of this paper is the following ontological inversion:

the particle has a position $\longrightarrow$ position-phase energy appears as a particle

The standard order is “particle $\rightarrow$ position $\rightarrow$ wave function $\psi(x) \rightarrow |\psi(x)|^2$,” whereas in this paper's rereading it is “finite-width position-phase-energy window $\rightarrow$ particle $\rightarrow$ observed wave form.”

Interaction between particles: an overlap indicator

Placing two particles 1, 2 as rectangular phase-energy windows, we introduce an indicator of the overlap region between the windows:

$$I_{12}(\theta) = R_1(\theta) \cdot R_2(\theta).$$

This is not the interaction itself, but an indicator of the region in which interaction can arise.

The actual interaction energy must be defined separately as a functional involving phase difference, energy density, and coupling constants. Depending on whether the value of $I_{12}(\theta)$ is zero or non-zero, we obtain the following correspondence regarding the region of potential interaction:

Overlap state	Expected physical consequence (candidate interaction kernel)
No overlap	No interaction
Overlap with phase alignment	Candidate for attraction / bound state (formation of a composite particle)
Overlap with phase mismatch	Candidate for repulsion / exclusion
Boundary contact	Candidate for tunneling / leakage

We obtain a natural candidate for unifying — through the same geometric structure (contact, overlap, phase alignment of finite-width phase-energy windows) — the interference, tunneling, scattering, binding, and repulsion that are treated separately in standard theory. Quantitative derivation of actual interaction energy, scattering cross sections, and binding energies is beyond the scope of this paper and is positioned as a follow-up task.

Correspondence with the box potential

The one-dimensional particle-in-a-potential problems treated in Chapter 5 of Shimizu’s textbook split into two families, the infinite well and the finite box barrier:

$$V_{\text{well}}(x) = \begin{cases} 0 & 0 < x < L, \\ \infty & \text{otherwise,} \end{cases} \quad V_{\text{barrier}}(x) = \begin{cases} V_0 & 0 < x < L, \\ 0 & \text{otherwise.} \end{cases}$$

In the infinite well, the boundary conditions $\psi(0) = \psi(L) = 0$ generate standing waves $\psi_n(x) = \sin(n\pi x/L)$. In this paper’s rereading, these appear as standing-wave modes of a rectangular phase-energy window under observation-bandwidth limitation.

In the finite box barrier, boundary connection gives rise to reflection, transmission, and the tunneling effect. In this paper’s rereading, the tunneling effect is read not as “the mystery of a point particle passing through the wall” but as “the connection of a finite-width phase-energy structure to the outside via boundary conditions.”

The common structure across both — “finite-width region and boundary conditions” — is what this paper’s rectangular phase-energy window picture focuses on.

Conclusion

A particle is not a point object but a rectangular phase-energy structure with central phase θ_x and finite width $\Delta\theta_x$. The observed wave nature is the effective wave form obtained by low-passing that rectangular body in the observation bandwidth (truncating the Fourier series at low order). The ontological inversion — not “the particle has a position” but “position-phase energy appears as a particle” — is the core of this section.

More precisely: This paper does not deny the Born rule. Rather, it treats the position-phase window $P_x(\theta)$ as a shape function (not a probability density). When the observation system reads this shape function as the wave function $\psi(x)$ corresponding to a detection basis $\{|\varphi_a\rangle\}$, the standard quantum-theoretic inner-product rule

$$p(a) = |\langle \varphi_a | \psi \rangle|^2$$

then yields the probability density. That is, this paper’s ontological hierarchy is the three-step structure “rectangular phase-energy window $R_x(\theta)$ (body) $\rightarrow$ low-order Fourier partial sum $S_N(\theta)$ (observed wave form) $\rightarrow$ probability distribution via inner-product projection onto the detection basis,” with the Born rule localized at the final projection step. The term “rectangular soliton standing wave” is used not in the strict sense of solutions to integrable systems but as a convenient name for finite-width, shape-preserving phase-energy structures. The calculational results of standard quantum theory — wave functions, probability densities, scattering cross sections, spectra — remain valid as effective theory under this rereading. What is radical about this section is not changing the formulas of standard theory but replacing the ontological reading of particle, position, and wave function.

Summary

Through seven thought experiments, the following structure emerges:

1. **The wall of identification (I)** — Δ is not a resolution but the threshold of individuation. $L \geq \Delta$ is the prerequisite for distance measurement.
2. **Non-identifiability of the locus of fluctuation (II)** — Δ on the ruler side and Δ on the physical-quantity side are not distinguishable from observation data alone.

3. **Wave-number-form uncertainty relation (III)** — $\Delta x \Delta k \geq 1/2$ is a mathematical fact about the Fourier transform; $\hbar$ appears as a unit-conversion factor.
4. **Quantum correlation as a composite wave packet (IV)** — by reading it as the spatial decomposition of a single wave packet, correlations between spatially separated systems can be explained without assuming information transmission.
5. **The algebra of observables (V)** — observables obeying an uncertainty relation are different projections of the same wave packet. The area element in phase space (symplectic capacity) gives the geometric invariant of the constraint.
6. **Complex-phase-space representation of physical quantities (VI)** — observed values are real projections; the substance of a physical quantity can be read as a phase-closure structure over a complex phase space. Uncertainty, interference, entanglement, and quantization are different projections of the same structure.
7. **Particles and the box potential (VII)** — a particle is a rectangular phase-energy window with central phase and finite width; the observed wave form is a low-order Fourier partial sum. Interactions between particles, read by treating the overlap region $I_{12}(\theta) = R_1(\theta)R_2(\theta)$ as a candidate interaction kernel, gain a geometric entry for description. From “the particle has a position” to “position-phase energy appears as a particle” — the ontological inversion.

These are consistent with the frameworks of complex Hilbert space, observables, measurement, uncertainty, the one-dimensional particle, and the box potential introduced in Chapters 1–5 of Shimizu’s *Foundations of Quantum Theory*, and provide one reading offering a conceptual overview of quantum theory above them. This paper does not modify the mathematical predictions of standard quantum theory; within the scope of Chapters 1–5, it offers a conceptual rereading compatible with the existing calculational formalism. The possibility of extension to regions beyond Chapters 1–5 is reserved as an independent task.

Related Work

The central rereading of this paper — “a particle is not a point object but a rectangular phase-energy window with central phase θ_x and finite width $\Delta\theta_x$, and the observed wave form appears as the low-order Fourier partial sum of that window” — partially resonates with a long lineage of attempts to find alternative pictures of quantum theory. This paper offers no new mathematical predictions, but clarifying the relation to the following prior works is useful for conceptual organization.

(1) The de Broglie double-solution program, Madelung hydrodynamics, and the Bohm pilot wave

The early matter-wave hypothesis of de Broglie [2] included attempts to read the particle as a stable singular wave (singular wave / soliton) in spacetime, and this is carried into the present as the “double-solution program” [22]. In parallel, Madelung [29] gave a hydrodynamic formalism rereading the wave function as fluid density and velocity field, and Bohm [30] completed this as the dualistic picture “pilot wave + particle.” The “rectangular phase-energy window” of this paper shares direction with these lineages in rereading the duality of probability wave and material structure. However, this paper presents neither solutions of specific nonlinear equations nor hidden variables; it remains an ontological rereading within Shimizu’s textbook framework.

(2) The Skyrme model, the MIT bag model, and non-topological solitons

The lineage of reading particles as “stable structures localized in a finite region of a field” includes Skyrme’s nonlinear field theory [23], the MIT bag model by Chodos et al. [24], and the non-topological soliton / Q-ball theory of Friedberg–Lee–Sirlin and Coleman [25][26]. These derive the finite-width structure of particles from specific Lagrangians in the contexts of the strong interaction and scalar field theory. The rectangular phase window of this paper shares with these attempts the intuition “particle = field structure in a finite region,” while presenting at a more geometric, formal level — reading the interaction $R_1(\theta) \cdot R_2(\theta)$ as an overlap in phase space.

(3) The Gabor transform, prolate spheroidal wave functions, and Hardy’s theorem

The operation of low-passing a rectangular phase window in the observation bandwidth runs mathematically parallel to the Gabor transform [36] and time-frequency localization framework in signal processing [27]. In particular, the prolate spheroidal wave functions (PSWF) of Slepian–Pollak [31] and Landau–Pollak [32] give rigorously the “family of functions most concentrated on a rectangular window in the time domain while band-limited to N ,” which is precisely the rigorous mathematical counterpart of this paper’s picture “rectangular body $R_x(\theta)$ + low-order Fourier partial sum $S_N(\theta)$.” Furthermore, Hardy [33] showed rigorously that a function ψ and its Fourier transform $\hat{\psi}$ cannot both decay rapidly at once; this provides the mathematical basis for our reading that “the rectangular body, at observation, can only be seen as a band-limited effective wave form.”

(4) Coherent states (minimum-uncertainty wave packets with central phase and finite width)

The most classical quantum-theoretic realization of a “phase-space structure with central phase and finite width” is the coherent state introduced by Schrödinger [34] in the harmonic-oscillator context and systematically formulated by Glauber [35] in quantum optics. Coherent states are Gaussian wave packets achieving the minimum uncertainty $\Delta x \Delta p = \hbar/2$, giving the standard quantum state with “center (x_0, p_0) and finite width” in phase space. The rectangular phase-energy window of this paper replaces the shape from Gaussian to rectangular and reads the observation image as a low-order Fourier partial sum — it can be seen as a radicalization of the coherent state. Both share the intuition “a localized quantum structure in phase space.”

(5) De Gosson quantum blobs and symplectic capacity

The central invariant of this paper — the area element in phase space — corresponds directly to the symplectic capacity originating from de Gosson’s quantum blobs [9] and Gromov’s non-squeezing theorem [15]. De Gosson–Luef [28] refines the geometric content of the uncertainty relation as symplectic capacity, and this paper’s rereading is consistent with that geometric framework.

Position of this paper

In relation to these prior works, the position of this paper can be summarized as follows: this paper presents no new dynamical equations or specific Lagrangians; it is an attempt, leaving the mathematical framework of Chapters 1–5 of Shimizu’s textbook unchanged, to replace the ontological reading of “particle,” “position,” and “wave function” on top of that framework. The main contribution is to show that the intuitions independently reached by lineages (1)–(5) — “particle = finite-width structure,” “observation = bandwidth limitation,” “minimum-uncertainty wave packet,” “phase-space area = quantum invariant” — emerge simultaneously on top of the elementary framework of Shimizu’s textbook.

Future Tasks

To develop the thought experiments of this paper into a full mathematical formulation, the following routes are conceivable:

1. **Formulating Thought Experiments I, II as POVM / coarse-grained measurement:** treat the wall of identification and the indistinguishability of fluctuation locus rigorously in the framework of Kraus operators and generalized measurement. Clarify the relation to decoherence theory when the apparatus Hamiltonian is known.
2. **Formulating Thought Experiment III as Fourier analysis / the Robertson inequality:** derive the conjugacy of position and wave-number from the Plancherel theorem and clarify the role of $\hbar$ as the dimension of action.
3. **Formulating Thought Experiment IV as tensor-product spaces / partial trace / the no-signalling theorem:** refine the composite-wave-packet picture as the geometry of reduced states of the density operator and quantitatively derive the maximum violation $2\sqrt{2}$ of the CHSH inequality (Tsirelson bound).

4. **Formulating Thought Experiment V as C*-algebras / von Neumann algebras / geometric quantum mechanics:** provide a self-contained description of the algebra of observables and clarify its correspondence to geometric quantum mechanics on the Kähler manifold structure of projective Hilbert space [11].
5. **Making explicit the correspondences with de Gosson’s quantum blobs [9] / the Wigner function [13] / the Moyal bracket [14]:** connect symplectic capacity, Wigner pseudo-probability distribution, and Moyal star product to phase-space quantum mechanics.
6. **Mathematical refinement of the complex-phase-space model (Thought Experiment VI):** specify the correspondences with the Wigner–Moyal formalism, deformation quantization, and Kähler quantum mechanics, and organize the relation to the experimental distinction between real-number and complex-number quantum theory by Renou et al. [21].
7. **Formulating the rectangular phase-energy window model (Thought Experiment VII):** refine the relation between the finite-width structure of the particle and the Fourier-partial-sum observation image in the language of density operators and spectral decomposition. Examine whether scattering cross sections, binding energies, and tunneling transmission can be derived from the overlap-region model $R_1(\theta) \cdot R_2(\theta)$.

These exceed the observational framework of this paper and should be treated as independent follow-up papers.

Priority outlook: Among the above, the most natural follow-up candidates are (1) formulating Thought Experiments I, II as POVM / coarse-grained measurement; (5) making explicit the correspondence with de Gosson’s quantum blobs; and (7) formulating the rectangular phase-energy window model. (1) rigorizes our starting point in direct connection to standard quantum measurement theory; (5) positions our core picture (the phase-space area element) within the existing symplectic-geometry framework; (7) opens the way to refining the newest claim of this paper — the finite-width structure of the particle — via density operators and spectral decomposition. With these as foundation, (2)(3)(4)(6) can be extended sequentially.

Acknowledgments

These thought experiments were organized step by step through AI dialogue. Verbatim records including the trial-and-error process are publicly available in the ai-chat-logs-open repository (/ (6)*, (7)*, (8)*, (9)*). The figures inserted in this paper are outputs of Python/matplotlib scripts implemented and executed in Claude Code from prompts generated in the AI dialogues (`figures/phase_position_wavepacket.py`, `figures/phase_window_body_and_observation.py`).

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